OPERATIONS & SUPPLY CHAIN

Designing the Flow, Making the Trade-offs

STUDENT PRE-READ GUIDE

A practical field guide to the decisions behind customer promises, operating performance, sourcing, fulfilment and resilient growth.

THE COURSE IN ONE LINE Customers experience one promise; the business must coordinate many connected decisions to deliver it.

How to use this guide

- Read the overview before Session 1, then read the relevant session brief before each class.
- You do not need to memorise formulas. Focus on the logic, evidence and decision each concept supports.
- Bring one example from a business you know. Ask what the customer is promised, what must happen behind the scenes, and where the system could fail.
- Questions are preparation prompts, not model answers. Be ready to explain your reasoning and the trade-offs you would accept.

Mesa PGP | Seven teaching sessions + one end-term assessment session

PART I | THE COURSE PRE-READ

The promise is visible. The operating system is not.

A customer taps “Order now” and expects a product to arrive on time, in the right condition, at a price that feels fair. That promise may depend on a forecast, supplier capacity, inventory placement, warehouse processes, software, delivery partners, route density, payment terms and a plan for what happens when something goes wrong. The customer sees one experience; the company runs a system.

Operations and supply chain management is the work of designing that system—and improving it as demand, costs, technology and expectations change. The challenge is not to make every activity as fast or as cheap as possible. It is to make the end-to-end system perform well.

A USEFUL MENTAL MODEL Promise → Plan → Source → Make/Prepare → Move → Deliver → Learn. Each step affects the others; hand-offs matter as much as individual activities.

Why local optimisation can make the whole system worse

Imagine a fulfilment centre automates picking and doubles the speed of that step. If packing is now the constraint, work-in-progress piles up, orders wait longer and the customer sees little improvement. Or a buyer negotiates a lower unit price by increasing the minimum order quantity. Purchase price falls, but cash is tied up in inventory and obsolescence risk rises. A decision can look excellent in one department and still damage the business.

Five lenses for every operating decision

Customer promise: What has the customer been led to expect—and what happens if we miss it?

Flow and time: Where does work wait, queue, repeat or get stuck?

Economics: What is the total cost after inventory, quality, service and failure costs?

Capacity and uncertainty: What changes when demand spikes, supply is late or processing time varies?

Resilience and options: What alternatives exist when the plan is wrong or a critical node fails?

A small toolkit—not a collection of formulas

- Map the process before proposing a fix. Show the work, decisions, hand-offs, queues and information needed to complete an order.
- Separate touch time from elapsed time. The work may take minutes while the customer waits hours or days.
- Find the constraint. Improving a non-constrained step may add cost without increasing total throughput.
- Use data to make uncertainty visible. A forecast is an input to a decision, not a promise about the future.
- Compare total cost and consequences. Unit price is only one component of cost.
- Make trade-offs explicit. Speed, cost, availability, flexibility and resilience rarely improve together for free.

The course journey

- 1. From Customer Promise to Operating Flow — where is time lost, and what is the real constraint?
- 2. Forecast the Future: Demand, Capacity & Inventory — how should we plan when the forecast is uncertain?
- 3. Improve the System: Bottlenecks, Queues, Variability & Process Improvement — how do we improve the whole system?
- 4. Source to Operate: Procurement as an Operating Lever — which sourcing choice creates the best total outcome?
- 5. Marketplace & Last Mile: Managing a Living Network — how do we balance supply, service and unit economics?
- 6. Design the Operating Model & Network — what footprint, capabilities and inventory locations should we build?
- 7. Resilience, Digital Visibility & Integrated Supply Chain Strategy — how can the system perform through shocks and scale?
- 8. End-term group presentations — can we defend an integrated recommendation with evidence?

Opening challenge

A D2C brand promises delivery within 24 hours. Orders are growing, customer complaints are rising, inventory is unevenly distributed, and the team is considering a new warehouse system and a cheaper supplier. Before choosing either investment, write down:

1. What exactly is failing from the customer's point of view?
2. What data would distinguish a capacity problem from a process, inventory or supplier problem?
3. Which decision could improve one metric while making another worse?
4. What is the smallest useful test to run before making a large investment?

KEEP THIS CHALLENGE IN MIND The sessions progressively add the tools needed to answer these questions with more structure and evidence.

SESSION 1

From Customer Promise to Operating Flow

THE BIG QUESTION Where does an order actually lose time—and which intervention would improve the customer promise without simply moving the problem elsewhere?

WHAT YOU SHOULD LEAVE WITH Map a process, distinguish work from waiting, identify the true constraint and propose a high-leverage redesign.

Why this matters

A delivery promise is the output of a chain of activities. In quick commerce, an order may move through acceptance, inventory confirmation, picking, packing, rider assignment and delivery. Every hand-off introduces the possibility of delay, missing information, rework or a queue. A process map makes those dependencies visible.

Concepts to know

- SIPOC gives a high-level view of Suppliers, Inputs, Process, Outputs and Customers.
- Swimlane maps show who or what performs each step and make hand-offs visible.
- Touch time is active work; waiting time is time spent idle or queued; lead time is elapsed time from start to finish.
- A bottleneck or constraint limits overall flow under current conditions. The busiest-looking step is not automatically the constraint.
- Little's Law: in a stable flow, WIP = Throughput × Flow Time. It links work in progress to how quickly work moves through.
- Redesign levers include eliminating unnecessary work, shifting work, automating a suitable step or decoupling activities.

Mini-case

A dark store receives an order at 7:00 pm. Inventory confirmation takes 1 minute, picking 4 minutes, packing 2 minutes, rider assignment wait 5 minutes and delivery 6 minutes. Total elapsed time is 18 minutes. If picking is reduced by a minute, the order may still miss a 15-minute promise because rider assignment and travel dominate. Diagnose across many orders, not one anecdote.

TRY IT BEFORE CLASS Sketch the order journey. Mark every hand-off, queue and possible rework loop. Which step would you investigate first, and what evidence would change your mind?

Questions to bring to class

- What is the difference between making a step faster and making the customer journey faster?
- When can a process map be misleading?
- How might a company improve one step while increasing total lead time?

Key takeaway

Do not start with a solution. Make the flow visible, then locate the delay or constraint that matters to the customer.

Suggested pre-class resources

- Watch/read (short): Jim Womack on why value-stream mapping matters

SESSION 2

Forecast the Future: Demand, Capacity & Inventory

THE BIG QUESTION When demand is uncertain, how should an operator use forecast evidence to decide capacity, inventory and the customer promise?

WHAT YOU SHOULD LEAVE WITH Interpret demand patterns, evaluate a simple forecast and connect uncertainty to capacity, inventory and service.

Why this matters

Businesses forecast because purchasing, staffing and inventory decisions must be made before actual demand is known. A forecast will be wrong; the practical questions are how wrong it tends to be, whether it is biased, how quickly it reacts to change and what the business should do about the uncertainty.

Concepts to know

- Demand patterns include trend, seasonality, promotions, intermittent demand and new-product uncertainty.
- Moving averages smooth recent observations; longer windows are smoother but slower to react. Weighted averages and exponential smoothing give selected or recent observations more influence.
- Forecast error is actual demand minus forecast demand. Bias asks whether forecasts are systematically too high or too low.
- MAE measures average absolute error. MAPE is intuitive but behaves poorly when actual demand is zero or very small; WAPE can be useful for aggregated demand.
- Capacity is the work a resource can handle over a period. Safety stock protects availability but consumes cash and space.
- S&OP / IBP connects demand, supply, inventory, finance and customer commitments.

Mini-case

A retailer planned for 10,000 units during a promotion; actual orders reach 17,000. Some products sell out early while others remain in stock. Expediting is available at 40% extra cost. The team must decide which orders to protect, whether to expedite, how to allocate inventory and what promise to make next. A forecast revision alone will not resolve the immediate operating choices.

TRY IT BEFORE CLASS If the next promotion is expected to generate 15,000 orders but the plausible range is 11,000–19,000, what would you flex, what would you hold fixed, and what trigger would make you change the plan?

Questions to bring to class

- When is a simple forecast preferable to a sophisticated model?
- Why can good average forecast accuracy still produce poor service?
- What should be decided before demand arrives, and what can wait for better information?

Key takeaway

Good planning does not eliminate uncertainty. It makes uncertainty visible and prepares decisions for more than one possible future.

Suggested pre-class resources

- Quick reference: MIT capacity lecture (Little's Law, queues and variability)

SESSION 3

Improve the System: Bottlenecks, Queues, Variability & Process Improvement

THE BIG QUESTION How do you improve total system throughput and responsiveness without simply shifting the bottleneck or increasing hidden WIP?

WHAT YOU SHOULD LEAVE WITH Diagnose queues and variability, assess system-wide interventions and explain when standardisation or automation is justified.

Why this matters

When a process operates close to full utilisation, small fluctuations in arrivals or processing times can create disproportionately large queues. The system may look productive because every resource is busy, yet orders spend more time waiting. Some spare capacity can improve total flow.

Concepts to know

- Utilisation is the share of available capacity being used; high utilisation is not always high effective throughput.
- Queues can be physical, digital or invisible, such as cases waiting for approval.
- WIP and Little's Law: with stable throughput, more WIP generally means longer flow time.
- Arrival variability comes from uneven demand; process variability comes from uneven work times.
- Batching reduces setup effort but can increase waiting. Push releases work based on a plan; pull responds to downstream need or capacity.
- Standard work and Lean tools reduce avoidable variation and make problems visible.
- Automation helps when it removes a genuine constraint; it can hurt when it accelerates a poorly designed process or moves the queue downstream.

Mini-case

A company automates picking. Pick rates rise, but packing and dispatch do not change. WIP accumulates between picking and packing; urgent orders become harder to find and late deliveries increase. Possible responses include redesigning packing, changing release rules, balancing labour, reducing batch size or limiting release into the next stage.

TRY IT BEFORE CLASS Before investing in another machine, name three pieces of evidence you would request. Include one measure of flow, one of variability and one of customer impact.

Questions to bring to class

- Why can reducing average processing time fail to reduce total lead time?

- When does batching help—and when does it hurt?
- How would you distinguish insufficient capacity from poor release rules or variability?

Key takeaway

Improve the system around the constraint. A faster local step is valuable only if it improves end-to-end outcomes.

Suggested pre-class resources

- Optional technical note: MIT queueing systems and Little's Law

SESSION 4

Source to Operate: Procurement as an Operating Lever

THE BIG QUESTION When the cheapest supplier creates operational pain, what does best value actually mean?

WHAT YOU SHOULD LEAVE WITH Compare sourcing options using total cost, operational fit, risk and flexibility—not unit price alone.

Why this matters

Supplier choices influence the whole operating model. Lead time affects replenishment and safety stock; minimum order quantities affect working capital; quality affects rework and returns; capacity affects responsiveness; payment terms affect cash flow. Procurement is not just a negotiation over price.

Concepts to know

- Total cost of ownership (TCO) includes purchase price plus relevant freight, inspection, defects, downtime, inventory and administration costs.
- Should-cost estimates what a product or service ought to cost based on cost drivers and assumptions; it is a discussion tool, not an unquestionable truth.
- Make vs buy compares internal capability and control with supplier economics, flexibility, risk and strategic importance.
- MOQ is minimum order quantity. A lower unit price tied to a large MOQ may increase cash tied up and obsolescence.
- Long or unreliable lead time may require buffers or weaken the customer promise.
- Supplier risk includes concentration, financial health, geography, capacity and availability of alternatives.

Mini-case

Supplier A quotes ₹100 per unit, requires an MOQ of 5,000 and has a variable four-week lead time. Supplier B quotes ₹108, allows orders of 1,000 and delivers reliably in one week. Price-only comparison favours A. If demand is uncertain and stock-outs are costly, B may create better economics—but calculate the relevant costs rather than assuming so.

TRY IT BEFORE CLASS List the costs that should be included before comparing these suppliers fairly. Which additional fact could reverse your recommendation?

Questions to bring to class

- Which costs are often omitted from supplier comparisons?
- When might a second source be worth paying more for?
- How can commercial terms change operational performance?

Key takeaway

Best value is the best total operating outcome under the conditions the business faces—not necessarily the lowest quoted price.

Important assessment note

The first 30 minutes of Session 4 are reserved for the individual assessment covering Sessions 1–3. Revisit process mapping, flow time, forecasting, capacity/inventory decisions, bottlenecks, queues, variability and system improvement. Focus on applying concepts and explaining why a recommendation makes sense, rather than memorising definitions.

Suggested pre-class resources

- Further reading: Chopra, Supply Chain Management: Strategy, Planning, and Operation (selected chapters on sourcing and coordination)

SESSION 5

Marketplace & Last Mile: Managing a Living Network

THE BIG QUESTION How do you balance customer experience, supply availability and unit economics when the network is constantly changing?

WHAT YOU SHOULD LEAVE WITH Explain how marketplace liquidity, last-mile density, incentives and service promises interact.

Why this matters

Marketplaces coordinate multiple parties whose decisions change in real time. Customers place orders, providers choose whether to participate, and the platform sets prices, incentives and service rules. Demand can surge in one neighbourhood while supply sits elsewhere.

Concepts to know

- Supply-demand matching connects available capacity to the right order, time and location.
- Liquidity is the likelihood of finding a suitable match quickly; low liquidity can increase waiting, cancellations and poor experience.
- Last-mile density can reduce cost per delivery, but only when timing, geography and service requirements allow efficient grouping.
- Batching and routing can improve utilisation but increase delivery time or reduce predictability.
- Incentives affect participation, but also contribution margin and behaviour.
- Unit economics asks what remains after costs associated with serving a transaction. Growth can destroy value if each additional transaction has negative contribution with no path to improvement.
- Weather, traffic, regulation, safety and competitor actions change feasible choices.

Mini-case

At 7:30 pm, food-delivery demand rises sharply during heavy rain. Travel slows, restaurants take longer to prepare orders and competitors increase rider incentives. The platform can raise incentives, increase delivery estimates, limit promotions, prioritise nearby orders or batch deliveries. Each action changes several metrics.

TRY IT BEFORE CLASS Choose two metrics you would protect first. Name one metric that may worsen as a result. What information would you need before changing incentives or delivery promises?

Questions to bring to class

- Why is adding more supply not always the fastest or cheapest fix?
- When can batching improve economics but damage customer experience?
- What is the difference between revenue growth and healthy marketplace economics?

Key takeaway

Marketplace operations are dynamic. Balance service, supply participation and contribution economics rather than maximising one metric.

Suggested pre-class resources

- Further reading: Chopra, Supply Chain Management: Strategy, Planning, and Operation (distribution and network chapters)

SESSION 6

Design the Operating Model & Network

THE BIG QUESTION How should a growing company configure its operating model, inventory and fulfilment network to meet customer needs at an acceptable cost?

WHAT YOU SHOULD LEAVE WITH Compare network choices and make trade-offs among service, cost, control, capacity and scalability explicit.

Why this matters

A startup may initially serve customers from one location, outsource fulfilment and hold limited inventory. As demand grows across regions and channels, the same model can become slow or expensive. The company must decide which capabilities to own, which to outsource, where inventory should sit and how many fulfilment nodes are justified.

Concepts to know

- Operating model describes how people, processes, technology, partners and governance deliver the product or service.
- Network design determines the number, location and role of suppliers, plants, warehouses, stores, hubs and delivery nodes.
- Ownership may provide control and capability; outsourcing may provide scale or specialist expertise but introduces dependencies.
- Centralised inventory can reduce duplication but increase distance and delivery time. Distributed inventory may improve speed but increase total inventory and complexity.
- A feasible network needs labour, supplier capacity, transport and operating capability—not just attractive map locations.
- Scalability means handling more volume or reach without costs and complexity rising at the same rate.

Mini-case

A growing D2C brand serves the country from one warehouse. Regional fulfilment centres could improve speed but require extra inventory, leases, systems and management. A third-party logistics provider may reduce upfront investment but lower control and introduce service dependencies. Compare demand by region, order profile, transport cost, delivery promise, fixed cost, implementation time and downside risk.

TRY IT BEFORE CLASS Draw two alternative networks for the brand. For each, state what improves,

what becomes more expensive or risky, and what demand evidence would justify the change.

Questions to bring to class

- What changes when a company moves from one region to a national footprint?
- Which capabilities should a founder keep close, and which could be outsourced?
- How should inventory positioning reflect service expectations and demand uncertainty?

Key takeaway

Network design is a long-lived set of trade-offs. Design for the customer promise and the economics the business can sustain.

Suggested pre-class resources

- Further reading: Chopra, Supply Chain Management: Strategy, Planning, and Operation

SESSION 7

Resilience, Digital Visibility & Integrated Supply Chain Strategy

THE BIG QUESTION How can an end-to-end operating system remain effective when demand, suppliers, capacity, technology and external conditions change?

WHAT YOU SHOULD LEAVE WITH Combine flow, planning, sourcing, network design, visibility and risk into a practical operating strategy.

Why this matters

A highly efficient system may have little slack, few suppliers and tightly synchronised inventory. That can work under stable conditions but be fragile when a supplier fails, a route closes or demand changes. Resilience means absorbing a shock, adapting and recovering at an acceptable cost—not preventing every possible disruption.

Concepts to know

- Concentration risk is dependence on one supplier, location, platform, route, technology or critical person.
- Optionality means credible alternatives such as backup suppliers, flexible capacity, alternate routes or postponement.
- Scenario planning tests how the plan performs under plausible futures rather than relying on one forecast.
- Digital visibility is timely, trustworthy information about orders, inventory, capacity, suppliers and delivery performance.
- A control tower brings operational signals together for exception management; a dashboard alone does not create control.
- AI and automation can support detection, forecasting and prioritisation but need reliable data, ownership and escalation paths.
- Sustainability constraints such as emissions, packaging, waste and labour practices can change sourcing and network choices.
- Executive metrics connect customer outcomes, flow, economics, risk and implementation progress.

Mini-case

A company plans to enter two markets when its largest supplier signals a six-week delay. Freight costs rise, demand is above plan in one region and below in another, and capital expenditure is capped. The team must decide whether to reallocate inventory, qualify an alternate supplier, change launch sequence, revise the customer promise or invest in visibility. A strategy must prioritise actions, owners, triggers and measures—not just list initiatives.

TRY IT BEFORE CLASS Identify the two most consequential risks. For each, propose one preventive action, one response option and one leading indicator that would tell management to act.

Questions to bring to class

- Which risks deserve redundancy, and which are better managed through flexibility or fast recovery?
- What makes operational visibility actionable rather than merely informative?
- When should leaders accept higher cost for resilience?

Key takeaway

A strong operating strategy integrates performance, economics and resilience—and turns analysis into executable decisions.

Suggested pre-class resources

- Optional reference: MIT OpenCourseWare — operations management, uncertainty, supply-chain design and sustainable operations

PART II | HOW TO PREPARE AND SUCCEED

What strong work looks like

- Diagnose before prescribing: define the customer or business problem before jumping to a solution.
- Use evidence: distinguish facts, estimates and assumptions. If data is missing, state what you would need.
- Think end to end: show how a change in one step affects the rest of the system.
- Quantify the trade-off: even a rough calculation with transparent assumptions is better than an unsupported opinion.
- Make the recommendation executable: identify first actions, owners, triggers and measures of success.
- Communicate with discipline: lead with the decision and the reason, then support it with the few facts that matter most.

Assessment structure

- Individual assessment in Session 4 — 30%: 30-minute individual submission covering Sessions 1–3.
- End-term group presentation in Session 8 — 50%: five minutes per group, integrating the full course; scored in class.
- Online binary questions in Sessions 1–7 — 10%: five True/False or Yes/No questions per teaching session, submitted online.
- Attendance — 10%: tracked and scored by Mesa.

A useful structure for the end-term presentation

5. Problem and customer promise: what is not working, and why does it matter?
6. Diagnosis: what does the process, demand, supplier or network evidence show?
7. Recommendation: what should change across the operating system?
8. Trade-offs and risk: what improves, what gets worse, and what is the contingency?
9. Execution and measures: what happens first, who owns it, and how will the team know it worked?

FINAL PRINCIPLE The goal is not to use every framework. It is to use the right evidence and concepts to make a clear, defensible decision.

Compact glossary

Capacity: The amount of work a resource or system can handle over a period.

Throughput: The rate at which completed units leave a process.

Lead time / flow time: Elapsed time from the start to the completion of a process.

WIP: Work-in-progress: items inside a process but not yet completed.

Forecast bias: A tendency for forecasts to be systematically too high or too low.

Safety stock: Buffer inventory held to protect against uncertainty in demand or replenishment.

TCO: Total cost of ownership across relevant lifecycle and operating consequences.

MOQ: Minimum order quantity required by a supplier.

Contribution margin: Revenue remaining after relevant variable costs; the exact definition depends on the business.

Resilience: The ability to absorb, adapt to and recover from disruption at an acceptable cost.

Optional further reading

- Eliyahu M. Goldratt, The Goal — constraints and system-level improvement.
- Donella H. Meadows, Thinking in Systems — feedback loops, interdependence and unintended consequences.
- Sunil Chopra, Supply Chain Management: Strategy, Planning, and Operation — network, inventory and supply-chain trade-offs.
- Association for Supply Chain Management (ASCM) — introductory resources on planning, operations and supply-chain terminology.
- Lean Enterprise Institute — introductory explanations of value-stream thinking, flow and Lean improvement.

The session briefs are designed to be self-contained. External material is optional; frameworks help structure a decision, but context, data and trade-offs determine which decision is appropriate.

- Read (5–7 min): Understanding the Fundamentals of Value-Stream Mapping — Lean Enterprise Institute

Start with the first item; the second is optional. Focus on the ideas that help you reason about the session's decision—not on memorising terminology.

- Optional reference: MIT OpenCourseWare — Operations Management lecture notes (capacity and inventory)

Start with the first item; the second is optional. Focus on the ideas that help you reason about the session's decision—not on memorising terminology.

- Read: MIT OpenCourseWare — capacity and queueing lecture notes

Start with the first item; the second is optional. Focus on the ideas that help you reason about the session's decision—not on memorising terminology.

- Optional reference: MIT OpenCourseWare — operations management topics, including supply-chain design and inventory

Start with the first item; the second is optional. Focus on the ideas that help you reason about the session's decision—not on memorising terminology.

- Optional reference: MIT OpenCourseWare — operations management cases, including Webvan and supply-chain coordination

Start with the first item; the second is optional. Focus on the ideas that help you reason about the session's decision—not on memorising terminology.

- Read: MIT OpenCourseWare — Supply Chain Design lecture resources

Start with the first item; the second is optional. Focus on the ideas that help you reason about the session's decision—not on memorising terminology.

- Read: Lean Enterprise Institute — value-stream mapping as a system-wide improvement tool

Start with the first item; the second is optional. Focus on the ideas that help you reason about the session's decision—not on memorising terminology.